Pairs, Photons and Holes — Torsion Conservation as a Common Principle

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Contents

.1 Initial Question and Methodology

Three phenomena of modern physics show at first glance a structural similarity:

- A photon splits into two photons of lower frequency (parametric fluorescence, SPDC).
- An excited electron forms a bound pair with a remaining hole (exciton).
- Two electrons with opposite spin form a Cooper pair (superconductivity).

In all three cases a bound pair arises from a single excitation state — and in all three cases conservation quantities are exactly satisfied.

Methodology of this document

This document strictly separates two levels:
[STANDARD]: Established physics, experimentally secured, documented with primary sources. These statements are valid independently of FFGFT.
[FFGFT]: Interpretation within the FFGFT framework. These statements are FFGFT-specific and should be read as such.
The aim is to rigorously establish the structural analogy between the three systems — and then to specify precisely where the FFGFT interpretation goes beyond standard physics.

.2 System 1: Parametric Photon Splitting (SPDC)

[STANDARD] Established Physics of SPDC

Spontaneous Parametric Down-Conversion (SPDC) is a nonlinear optical process: a photon with frequency ω_0 is converted in a nonlinear crystal into two photons of lower frequency [**?**]:

$$\omega_0 \longrightarrow \omega_1 + \omega_2, \quad \text{with } \omega_1 + \omega_2 = \omega_0. \tag{1}$$

Exactly two conservation laws hold:
Energy conservation:

$$\hbar\omega_0 = \hbar\omega_1 + \hbar\omega_2. \tag{2}$$

Momentum conservation (phase matching):

$$\mathbf{k}_0 = \mathbf{k}_1 + \mathbf{k}_2. \tag{3}$$

For type-II SPDC the resulting photons have **opposite polarisation** and are **entangled**. The total angular momentum of the system is conserved [**?**].

STANDARD: What SPDC directly shows

1. A higher-energy state splits into two lower-energy states.
2. Energy and momentum are exactly conserved.
3. The opposite polarisation of the photons is not an additional assumption — it follows from angular momentum conservation.
4. The resulting photons are **real photons** — not quasi-particles, not computational aids. *[FFGFT caveat: see below — photons in FFGFT are not point particles but extended wave packets.]*

[FFGFT] Interpretation of SPDC

FFGFT: SPDC as torsion splitting — photons as wave packets

In FFGFT every photon carries angular momentum as a topological property of the electromagnetic field torsion (Doc. 174, Section 3).
Important FFGFT basic position: photons are wave packets.
In standard theory (QED), the photon is a quantised excitation quantum of the electromagnetic field — mathematically a Fock state $|1_{\mathbf{k}}\rangle$. It is often treated as a point particle.
In FFGFT the photon is **not a point particle**, but an **extended wave packet**: a spatially bounded, coherent torsion wave of the electromagnetic field with defined frequency ω, wave vector **k** and polarisation.
Why this matters for SPDC: The splitting $\omega_0 \to \omega_1 + \omega_2$ is in FFGFT not the decomposition of a point particle, but the **splitting of a wave packet** into two wave packets of lower frequency in the nonlinear crystal. This is physically more direct — light *is* a wave, and the splitting is a wave process.

Standard theory (QED)	FFGFT
Photon = Fock state $|1_{\mathbf{k}}\rangle$	Photon = extended torsion wave packet
Point particle picture for many calculations	No point particle; always wave packet
Splitting = annihilation + creation of quanta	Splitting = wave process in non-linear medium
Entanglement = quantum correlation of Fock states	Entanglement = topological torsion correlation

Consequence for the description "real photon": "Real" means in FFGFT: no quasi-particle approximation, no many-body reformulation — but no point particle. A SPDC photon is a real extended wave packet of the torsion field with clearly defined properties (ω, **k**, polarisation, spatial extent).
Conserved quantity in FFGFT: The topological torsion charge is additive and conserved. This is no new claim — it is the FFGFT formulation of the known angular momentum conservation, now formulated for wave packets instead of point particles.

.3 System 2: The Exciton — Electron and Hole

[STANDARD] What a Hole Really Is

Here lies the conceptual core.

STANDARD: The hole is not a real particle

A **hole** in the valence band of a semiconductor is **not an independent fundamental particle** — it is a **quasi-particle** that describes the collective reaction of the many-body system to a missing electron [**?**, **?**].
Precisely formulated (standard textbook definition):

- The valence band is completely filled in the ground state.

- When an electron is missing (through absorption of a photon), the many-body system behaves *as if* a positively charged particle with positive effective mass were present.
- This fictitious particle is called a "hole".
- It has: positive charge $+e$, effective mass m_h^*, and a spin opposite to that of the missing electron.

The hole is mathematically equivalent to the description in electron language — but it is a construct of the many-body approximation, not a real antiparticle (not a positron!).

The real antiparticle of the electron is the **positron** — which arises in pair production ($\gamma \to e^- + e^+$) from real high-energy photons. The hole in the semiconductor is structurally different: it exists only in the context of the valence band of the material.

[STANDARD] The Exciton

The **exciton** is a bound state consisting of an excited electron in the conduction band and a hole in the valence band [**?**, **?**]:

$$\text{Exciton} = \text{Electron (conduction band)} + \text{Hole (valence band)}. \tag{4}$$

The binding arises through the Coulomb attraction between the negatively charged electron and the positively charged hole.

Two main types:

- **Wannier-Mott exciton**: weakly bound (binding energy 1–100 meV), spatially extended (radius $\gg a_{\text{lattice}}$). Typical for semiconductors (GaAs, CdSe, Si). Stable only at low temperatures.
- **Frenkel exciton**: strongly bound (binding energy 0.1–1 eV), spatially compact (radius $\sim a_{\text{lattice}}$). Typical for organic semiconductors, biological systems. Stable even at room temperature.

[FFGFT] Interpretation of the Exciton

FFGFT: Exciton as complementary torsion pair

In FFGFT the electron in the conduction band carries a definite torsion configuration (spin, ξ-level N). The hole in the valence band is described as anti-torsion: the absence of an electron with opposite quantum numbers.

The exciton is in FFGFT a **torsion-anti-torsion pair**: the total torsion is reduced (Spin S_{total}, Momentum $\mathbf{p}_{\text{total}}$ are conserved), analogously to SPDC.

Critical limitation: The hole is a quasi-particle in standard theory — no fundamental particle. The FFGFT description as "anti-torsion" is an interpretation that goes beyond standard physics and has not been independently proven.

.4 System 3: The Cooper Pair

[STANDARD] BCS Theory and the Cooper Pair

In the BCS theory of superconductivity [?, ?], two electrons with opposite spin and momentum form a bound pair:

$$\text{Cooper pair} = \text{Electron}(\mathbf{k},\uparrow) + \text{Electron}(-\mathbf{k},\downarrow). \tag{5}$$

The binding is not direct (two negative charges repel each other) but is mediated via phonons (lattice vibrations):

- Electron 1 polarises the crystal lattice.
- This polarisation attracts Electron 2 with a delay.
- The result is an indirect, weak attractive interaction.

Binding energy: $\Delta E \approx 1$–10 meV (conventional superconductors). This is much smaller than $k_B T$ at room temperature (≈ 26 meV).

Type	T_c	Binding energy	Mechanism
Conventional	1–40 K	1–10 meV	Phonon-mediated
High-T_c	up to 138 K	~ 50–100 meV	Not fully understood

[FFGFT] Interpretation of the Cooper Pair

FFGFT: Cooper pair as opposite torsion configurations

In FFGFT the two electrons of the Cooper pair have opposite torsion configurations: $(\mathbf{k},\uparrow)$ carries torsion in one direction, $(-\mathbf{k},\downarrow)$ carries torsion in the opposite direction. The pair thus has:

- Total torsion zero (spin singlet)
- Total momentum zero ($\mathbf{k} + (-\mathbf{k}) = 0$)

This is a **topologically neutral state** — geometrically more stable than two free, separate torsion configurations.

Why low temperature is necessary (FFGFT interpretation): The binding does not arise directly from the torsion geometry, but is mediated via phonons (lattice vibrations). Phonons are collective vibration modes of the crystal lattice — they lie at a different ξ-level than the electron spin system. The phonon-mediated coupling is therefore a cross-level coupling between ξ-levels (Doc. 175, section on ξ-hierarchy), whose strength $\propto \xi^{\Delta N}$ with $\Delta N = 1$. This explains the weakness of the binding energy compared to direct spin-spin coupling.

Critical limitation: This interpretation is consistent but not proven. Standard BCS theory explains the same situation completely without FFGFT concepts.

.5 Comparison of the Three Systems

[STANDARD] Structural Commonalities

System	Constituents	Binding energy	Temperature	Conserved
SPDC	Photon 1 + Photon 2	— (no binding)	Room temperature	E, $\mathbf{p}$, J_z
Exciton	Electron (conduction band) + Hole (valence band, quasi-particle)	10–100 meV (Wannier) to 1 eV (Frenkel)	4 K–300 K depending on type	E, $\mathbf{p}$, S, charge
Cooper pair	Electron 1 $(\mathbf{k}, \uparrow)$ + Electron 2 $(-\mathbf{k}, \downarrow)$	1–10 meV (conventional)	$< T_c$ (1–40 K)	E, $\mathbf{p} = 0$, $S = 0$

Structural commonality [STANDARD]: In all three cases, two objects with opposite quantum numbers (spin, momentum, polarisation) arise from an excited state, so that conservation quantities are exactly satisfied.

Critical difference [STANDARD]:

- **SPDC:** two real photons, no binding energy, no temperature limit.
- **Exciton:** real electron + quasi-particle (hole), Coulomb binding, temperature-sensitive (Wannier-Mott: $T \ll T_{\text{bind}}/k_B$).
- **Cooper pair:** two real electrons, phonon-mediated binding, very temperature-sensitive ($T < T_c$).

The hole in the exciton is **not** analogous to the second photon in SPDC — it is a quasi-particle without independent fundamental existence outside the many-body system.

[STANDARD] The User's Question: "Hole = Two Electrons?"

This is a precise and important question.

The answer is: **technically yes, conceptually no**.

Technically yes: The exciton description "electron + hole" is mathematically equivalent to a description exclusively in electrons. In many-body theory the hole is defined as:

$$|\text{hole}\rangle \equiv |\text{Fermi sea without 1 electron at } \mathbf{k}, \downarrow\rangle . \tag{6}$$

The hole *is* the missing electron in disguise.

Conceptually no: The hole description is not merely a renaming — it is an efficient reorganisation of the many-body states that describes many phenomena (conductivity, magnetotransport, optical transitions) much more simply than pure electron language. The hole *behaves* like a particle — it has an effective mass, a spin, a momentum. It is a useful construct with real physical content, even if it is not a fundamental particle.

Hole: useful construct, not a fundamental particle

The hole in the semiconductor is not:

- A real antiparticle (not a positron)
- A new fundamental structure of nature
- Independent of the electron description

The hole is:

- A quasi-particle: an emergent excitation of the many-body system
- Mathematically: the absence of an electron in the valence band, rewritten as a positively charged particle
- Physically meaningful: it correctly describes real, measurable phenomena

The **Cooper pair**, by contrast, consists of **two real electrons** — no hole, no quasi-particle. That is the fundamental difference.

.6 [FFGFT] The Common Principle: Torsion Conservation

FFGFT: Conservation of total torsion as a unified principle

In FFGFT the connecting principle behind all three systems is the **conservation of total torsion**:

System	Before the process	After the process	Conserved
SPDC	Torsion ω_0, angular momentum J_z	Torsion ω_1 (+) + Anti-torsion ω_2 (–)	Torsion charge: $J_z = J_1 + J_2$
Exciton	Valence band torsion (full band)	Conduction band torsion (electron) + valence band anti-torsion (hole)	Torsion charge: S_{total}, $\mathbf{p}_{\text{total}}$
Cooper pair	Fermi sea torsion	Pair torsion $(\mathbf{k}, \uparrow)$ + $(-\mathbf{k}, \downarrow)$	Torsion charge: $S = 0$, $\mathbf{p} = 0$

The common FFGFT language: In all three cases a pair of torsion and anti-torsion arises from a torsion configuration, such that the total torsion charge is conserved. This is in FFGFT no coincidence — it is the mirror image of the known conservation laws (energy, momentum, angular momentum) in the language of the torsion field.

What FFGFT does not claim: FFGFT does *not* claim that exciton hole and Cooper electron and SPDC photon are the same objects — they are physically different. FFGFT claims that their *structural similarity* has a common geometric origin: the conservation of the topological torsion charge.

.7 Consistency with the Temperature Argument from Doc. 173

The three systems differ dramatically in their temperature stability. This is [STANDARD] and directly explicable:

System	Binding energy	Stability temperature	Cause
Photon (SPDC)	none (free photons)	Room temperature	No thermal dissipation
Frenkel exciton	0.1–1 eV	Room temperature possible	$\Delta E \gg k_B T(300\,\text{K}) = 26\,\text{meV}$
Wannier exciton (CdSe)	~ 15–20 meV	$< 50\,\text{K}$	$\Delta E \sim k_B T$ at room temperature
Cooper pair (conventional)	1–10 meV	$T < T_c = 1$–40 K	$\Delta E \ll k_B T(300\,\text{K})$

Consistency with Condition 2 from Doc. 173

The temperature limits follow directly from Condition 2 (Doc. 173): $\Delta E > k_B T$. The weaker the binding energy, the lower the temperature must be — this is no FFGFT argument, this is standard physics.
The Cooper pair needs low temperature not because of its electron nature, but because of its **very weak binding energy**, which follows from the indirect phonon mechanism.

.8 Summary

Conclusion: What is certain and what is FFGFT interpretation

[STANDARD] Established statements:

1. The hole is a quasi-particle, not a real antiparticle.
It is the mathematical reformulation of a missing electron in the valence band. “Exciton consists of two electrons” is technically correct — but the hole description is more efficient and physically meaningful.

2. The Cooper pair consists of two real electrons.
Opposite spin and momentum. Phonon-mediated binding. Low temperature is necessary because of the weak binding energy ($\Delta E \ll k_B T_{\text{room temperature}}$).

3. SPDC produces two real photons — in FFGFT: two wave packets.
Energy, momentum and angular momentum are exactly conserved. No binding energy — the photons are free. In FFGFT photons are not point particles but extended torsion wave packets — the splitting is a wave process, not the decay of a point particle.

4. The structural similarity is real:
In all three cases two objects with opposite quantum numbers arise from an initial state, so that conservation quantities are exactly satisfied.

[FFGFT] Interpretation:

5. Torsion conservation as a unified principle.
The known conservation laws (energy, momentum, angular momentum) correspond in FFGFT to the conservation of the topological torsion charge. The structural similarity of the three systems thus has a common geometric origin — without the three systems being the same.

6. The hole as anti-torsion.
Consistent with the FFGFT description of the exciton from Doc. 173 — but going beyond standard physics and not independently proven.

Bibliography

[1] Burnham, D. C. & Weinberg, D. L. *Observation of simultaneity in parametric production of optical photon pairs.* Phys. Rev. Lett. **25**, 84–87 (1970).

[2] Kwiat, P. G. et al. *New high-intensity source of polarization-entangled photon pairs.* Phys. Rev. Lett. **75**, 4337–4341 (1995).

[3] Frenkel, J. *On the transformation of light into heat in solids. I & II.* Phys. Rev. **37**, 17–44 and 1276–1294 (1931).

[4] Wannier, G. H. *The structure of electronic excitation levels in insulating crystals.* Phys. Rev. **52**, 191–197 (1937).

[5] Kittel, C. *Introduction to Solid State Physics.* 8th ed. Wiley (2004). Chapter 8: Semiconductor Crystals.

[6] Ashcroft, N. W. & Mermin, N. D. *Solid State Physics.* Holt, Rinehart and Winston (1976). Chapter 12: The Semiclassical Model of Electron Dynamics; Chapter 28: Homogeneous Semiconductors.

[7] Cooper, L. N. *Bound electron pairs in a degenerate Fermi gas.* Phys. Rev. **104**, 1189–1190 (1956).

[8] Bardeen, J., Cooper, L. N. & Schrieffer, J. R. *Theory of superconductivity.* Phys. Rev. **108**, 1175–1204 (1957).

[9] Woggon, U. *Optical Properties of Semiconductor Quantum Dots.* Springer Tracts in Modern Physics, Vol. 136 (1997).